

Performance Comparison Report: Titanic Dataset

Executive Summary: This report outlines the predictive modeling efforts on the Titanic survival dataset. Four classification algorithms were evaluated, alongside feature engineering and hyperparameter tuning, to identify the optimal model for survival prediction.

1. Dataset Description & Preprocessing

The dataset used is the well-known **Titanic survival dataset**. It comprises historical data of passengers, with the target variable being Survived (binary classification).

- **Dataset Split:** The data was partitioned into a training set (623 records, 16 features) and a test set (268 records, 16 features).
- **Data Cleaning:** Missing values in key columns like Age and Embarked were imputed, while columns with excessive missing data (e.g., Cabin) were considered for exclusion. Outliers in the Fare column were also identified and assessed.
- **Feature Engineering:** New informative features were derived from existing data, including generating a family_size feature from SibSp and Parch, and extracting socio-economic titles from the Name column to improve model signal.

2. Models Used & Evaluation Metrics

The predictive modeling phase involved training four different baseline algorithms to establish performance benchmarks:

- **Logistic Regression:** A foundational linear model for binary classification.
- **Decision Tree Classifier:** A non-linear, tree-based model useful for capturing complex decision boundaries.
- **Random Forest Classifier:** An ensemble approach utilizing multiple decision trees to reduce variance and combat overfitting.

- **XGBoost Classifier (Extreme Gradient Boosting):** A highly optimized, sequential ensemble method that builds trees to correct residual errors.

Evaluation Metrics: The primary metric used to evaluate and compare the models was **Accuracy**, supplemented by the analysis of True Positives, True Negatives, False Positives, and False Negatives via a Confusion Matrix logging in MLflow.

3. Comparison of Results

All models were initially evaluated without hyperparameter tuning to gauge their inherent performance on the dataset. The results of the baseline tracking (logged via MLflow) are summarized below:

Model	Baseline Accuracy
XGBoost Classifier	80.22%
Logistic Regression	79.85%
Random Forest Classifier	78.36%
Decision Tree Classifier	76.12%

4. Best-Performing Model & Optimization

The **XGBoost Classifier** emerged as the strongest baseline model, achieving the highest accuracy of **80.22%**. This highlights the algorithm's capability to effectively model complex, non-linear relationships and interactions within the passenger data.

Hyperparameter Tuning (Decision Tree Case Study)

While XGBoost had the highest baseline, hyperparameter tuning was successfully applied to the Decision Tree Classifier using RandomizedSearchCV to demonstrate performance gains. The model was tuned across hyperparameters including max_depth, min_samples_split, and min_samples_leaf.

- **Optimal Parameters Found:** max_depth: 5, min_samples_leaf: 5, min_samples_split: 4.

- **Performance Gain:** Accuracy improved significantly from **76.12% to 79.10%**, demonstrating the critical value of parameter optimization.

5. Key Findings and Recommendations

- **Tree-based Ensembles Excel:** As expected with tabular data, gradient boosting (XGBoost) outperformed both linear models and standard bagging ensembles (Random Forest) in its baseline state.
- **Feature Engineering Adds Value:** The creation of synthesized features like `family_size` and name titles proved crucial, providing strong predictive signals that enhanced the overall accuracy across all algorithms.
- **Recommendation for Deployment:** It is recommended to proceed with the **XGBoost Classifier** as the primary production model. However, performing a thorough hyperparameter tuning sweep (via GridSearch or RandomSearch) on this XGBoost model is highly advised to extract further performance gains before final deployment.